

Queensland University of Technology
Faculty of Science
School of Mathematical Sciences

PhD Stage 2 Proposal

Computational Modelling of Multilateral Treaties

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Proposed thesis title and type

“Computational Modelling of Multilateral Treaties”, thesis by publication.

Proposed supervisory team and their credentials

Role	Name	Affiliation
Principal Supervisor	Prof. Michael Bode	School of Mathematical Sciences, QUT
Associate Supervisor	Prof. Jonathan Rhodes	School of Biology and Environmental Science, QUT

Principal Supervisor — Prof. Michael Bode Prof. Bode is Deputy Director of Capability Development at SAEF (Securing Antarctica’s Environmental Future). His work centres on applied mathematics and conservation, with a focus on decision-making under uncertainty and mathematical biology. His previous work on Antarctic Treaty System (ATS) decision-making and conservation makes him an expert on consensus-based systems.

Associate Supervisor — Prof. Jonathan Rhodes Prof. Rhodes is Director of the Centre for Environment and Society. He specialises in spatial ecology, environmental governance, and the modelling of coupled human–natural systems, bridging quantitative models with real-world policy. His expertise gives him insight into the policy implications of environmental decisions, grounding the treaty-system applications of this research.

Chapter 1

Background and Literature Review

1.1 Introduction

Consensus is a collective decision-making process where every party can object to any proposal and effectively veto it. A body becomes a consensus regime when consensus is the only means to reach decisions. Consensus is not the same as unanimity: a unanimity rule puts a proposal to a vote and records who votes in favour, against or abstain, whereas under consensus a decision proceeds whenever no party registers a formal objection (Rietig et al. 2023). Since a single party can halt any proposal, an outside observer might expect such regimes to decide very little. The larger and more heterogeneous the group, the smaller the set of proposals that change the status quo (Tsebelis 2002). In practice, consensus regimes can reach agreement even on contested, transnational issues, often by relying on informal practices: trust, dialogue, and leadership. These practices fill the gaps left by their formal rules (Sykora-Bodie and Morrison 2019). But success is never guaranteed: in bodies such as the United Nations Security Council a single veto routinely blocks collective action, and even historically productive regimes can stall (Gardiner et al. 2025). Quantitatively, the picture is no different: what separates a consensus regime that keeps reaching decisions from one that can no longer reach them is not captured by existing models of consensus dynamics (Pérez et al. 2018; Tran et al. 2024). A model that better reproduces how consensus is reached or broken is therefore needed, both to explain past behaviour and to anticipate when a regime approaches the limits of its capacity to act.

Models of multi-actor decision-making already exist, but they were not built for this task: they represent each actor with a single, low-dimensional position and rarely use the actual text those actors produce. Capturing a consensus regime faithfully requires positions across many issues, read directly from the text the actors produce, and validated against observed outputs rather than assumed. Recent advances in natural

language processing and machine learning make each of these upgrades feasible.

The Antarctic Treaty System (ATS) is one of those consensus-driven regimes. At its centre is the Antarctic Treaty (AT), whose consultative parties meet yearly at the Antarctic Treaty Consultative Meeting (ATCM), where outputs are adopted only by consensus (Secretariat of the Antarctic Treaty 2023). In 2022, for the first time, the ATCM parties failed to reach consensus on the final report; even as activity has increased, the regime has become less responsive to challenges (Gardiner et al. 2025).

1.2 Modelling consensus: responsive and anticipatory approaches

There are two traditional groups of models for how groups reach consensus. The first group is responsive and interactive: each actor repeatedly reacts to the current positions of the others. These are the opinion-dynamics models. They treat consensus as *convergence*: opinions are iteratively averaged until they meet (DeGroot 1974), with several variants (Appendix E). Opinion-dynamics models can nevertheless test for agreement: declare consensus when all final positions lie within a tolerance radius, as Bernardo et al. (2021) do for the negotiation of the 2015 Paris Agreement. That test, however, captures preference convergence, not the consensus rule the ATS uses: a procedural right under which one party can block a decision no matter how close the remaining positions are.

The second group is anticipatory: each actor reasons about how the others will respond and chooses its actions to maximise an objective function. Responsive agents, by contrast, follow an update rule with no objective at all. These models use game theory to treat consensus as a unanimity rule: in the veto-players framework of Tsebelis (2002), the status quo is replaced only when no party is left worse off (formal condition in Appendix E). As parties multiply and their positions spread, the set of adoptable policies therefore shrinks. Analysing the ATS, Gardiner et al. (2025) point to the same ingredients: a growing and diversifying membership, divergent views, and a consensus rule, and read the regime's shift away from binding agreements as declining performance.

Each group has strengths and limits. Responsive models are simple, can be scaled to many agents, and are easy to calibrate against data. But their agents have no objective, no veto, and no strategy. Anticipatory models represent vetoes and strategic choice directly, and they explain *why* a proposal passes or fails. But their equilibria become hard to compute as parties and issues multiply, and they are rarely confronted with data from real negotiations.

The ATS is this thesis’s case study of such a regime: its Consultative Parties (CPs) decide exclusively by consensus at the ATCM, and its best-documented episode of contested consensus is CRAMRA, a minerals convention agreed in 1988 and then abandoned in favour of the Madrid Protocol’s mining ban (Jackson 2021). That documented reversal provides a canary test for any preference measure this thesis builds. The regime’s mechanics, outputs, and the full CRAMRA episode are described in Appendix F (Gardiner et al. 2025).

Between these two groups lies a middle ground: models that retain the strategic, veto-bearing structure of the second group but learn equilibrium from repeated interaction instead of solving for it. This field is treated in the next section.

1.3 Negotiation and coalition formation simulation using reinforcement learning

Section 1.2 left two requirements for a model of ATS behaviour. The model must represent many heterogeneous parties, each holding a veto and bargaining over a status quo. It must also remain computable at that scale, which the closed-form equilibria of analytic game theory are not. Multi-agent reinforcement learning (MARL) meets both requirements: it keeps the game-theoretic substance but learns equilibrium behaviour from repeated interaction instead of solving for it.

Formally, MARL is defined over stochastic (Markov) games, a generalisation of repeated games in which the state of the negotiation changes as the agents act. Each agent observes the state, chooses an action, and receives a reward: the objective function it learns to maximise. The learning procedures approximate equilibrium behaviour over many simulated rounds rather than deriving it in closed form (Littman 1994).

The field spans foundational bargaining and coalition-formation work (Gronauer and Diepold 2022; Sarkar et al. 2022) and, most recently, agents based on Large Language Models (LLMs) that model political negotiation directly (Appendix E). However, these simulations run on synthetic data or abstract benchmarks rather than the deliberative records of a real consensus regime.

1.4 Quantitative and text-based study of institutions and treaties

The quantitative study of international institutions long predates its application to Antarctica. One line of work codes the design of treaties and organisations as structured variables, such as their decision rules, and links these variables to institutional performance, survival, and decline (Gray 2018). A more recent trend treats the *text* of international agreements and debates as data (Grimmer and Stewart 2013), recovering state positions from voting records and speeches (Baturu et al. 2017; Slapin and Proksch 2008). Both show that the preferences and design features of treaty regimes can be measured, not merely described.

The ATS’s health has been assessed qualitatively (Mancilla and Jabour 2023) and quantitatively, in two ways. Gardiner et al. (2025) measures the regime in aggregate, counting inputs, outputs, and the time legally binding measures take to enter into force, and concludes that responsiveness to challenges in the Antarctic region is declining. The second way is structural. A Working Paper (WP), the document type through which parties bring proposals to the ATCM (Appendix F), may be submitted by a single party or co-sponsored by several, and this co-sponsorship has been used to map who collaborates with whom through co-authorship networks of ATCM documents (Botelho et al. 2026; Kumar et al. 2026). This thesis uses both lines of work as validation targets: parties with similar stances (directional positions on a topic) are likely to co-sponsor papers together, and the same measures should help predict the output series reported by Gardiner et al. (2025) (Section 2.1.2). Neither analysis, however, reads the content of the papers, so neither can say which issues divide the parties, a limitation taken up in Section 1.5.

A separate field does read the content: positional measures place each actor on a policy scale using only the text the actor produces. It has two paradigms, which differ in how they turn text into numbers: word counts (i.e. frequencies) or embeddings, vectors in which texts with similar meaning end up close together. The word-count paradigm recovers a single, uninterpreted dimension, which in a corpus as topically diverse as the ATCM Working Papers would likely reflect what parties talk about rather than what they support. This thesis therefore adopts the embedding paradigm: topics are discovered with BERTopic (Grootendorst 2022), and stances are read with natural language inference (NLI) classifiers, which judge whether a text supports or contradicts a stated hypothesis (Burnham 2023; Abercrombie and Batista-Navarro 2022). The result is a topic-conditioned stance: a separate position per party and topic through time. The supporting literature and definitions are reviewed in Appendix G, and Section 2.1.2 sets out the validation checks.

1.5 Summary of literature gap

The descriptive work of Section 1.4 (Gardiner et al. 2025; Mancilla and Jabour 2023) shows what the regime outputs but not who produces what. The structural analyses of co-authorship networks (Botelho et al. 2026; Kumar et al. 2026) are undirected and unattributed. The text-based positional methods of Section 1.4 recover latent positions rather than topic-specific stances over time (Appendix G). Finally, consensus has been simulated as convergence (Section 1.2) or on synthetic data (Section 1.3). These literatures differ in method but none recovers a directional preference: who is for or against what, extracted from the regime’s own words.

This gap has two parts. The knowledge gap is the question raised in Section 1.1: whether the regime’s outputs, including its failures, can be anticipated from the parties’ positions at all. The methodological gap is the missing instrument: no existing method measures each party’s directional stance, per topic and over time, from the regime’s own records. Without that instrument the question cannot be answered. Section 1.6 turns this gap into three research goals.

The closest prior work shares parts of this design but not its object. Stance detection recovers for-or-against positions from parliamentary debates (Abercrombie and Batista-Navarro 2022), but not a per-party, per-topic measure through time. Moghimifar et al. (2024) drive a negotiation simulation with preferences derived from party manifestos, but in a parliamentary coalition setting, not a consensus regime where every party holds a veto. No prior work extracts a directional preference for each party over time from a consensus regime’s own textual records, nor uses such preferences to drive a strategic simulation in which agents have veto power. This thesis supplies both halves and joins them: the preferences recovered from the record become the parameters of the simulation that reproduces consensus and its failures.

1.6 Research problem

The first goal of this thesis is to reconstruct parties’ preferences from the written records as a directional measure (stance) across topics and time. The second goal is to extract a directional opposition (veto) signal from the same records, and to test whether the combined preferences recover known history and predict decisions. Finally, this thesis aims to use these preferences to parameterise a multi-agent model that simulates consensus and its failures.

Chapter 2

Program and design of the research project

2.1 Objectives, methodology and research plan

This thesis has three interconnected goals. The first is to reconstruct each party's preferences from the regime's written records, as a directional measure across topics and time (Chapter 1). The second is to test whether the recovered preferences are credible: they must match known historical positions and align with the ATCM decision record (Chapter 2). The third is to use the validated preferences to simulate the regime's collective decision-making, so that its outputs, including consensus failures, can be anticipated (Chapter 3). Figure 2.1 shows the overall flow of the research project, the output of each stage, and how the objectives relate to one another.

2.1.1 Objective 1 / Chapter 1: The Support layer

Aim To produce a defensible, directional measure of parties' preferences over relevant topics through time, using documents the parties submit themselves. For the ATCM case, Working Papers (WPs) are the input data because, of all submitted documents, these are the only ones submitted by parties or Observers (non-Secretariat) that require discussion and action. This makes WPs the most likely text to contain parties' strategic intentions. They are also where outcomes originate: nothing happens without them, and Measures generally begin as WPs. The analysis uses the English-language WPs in the Secretariat's public database. WPs are translated into the four treaty languages (Secretariat of the Antarctic Treaty 2023), so the English corpus is not limited to papers originally drafted in English, and most pre-trained embedding and NLI models are trained on English data.

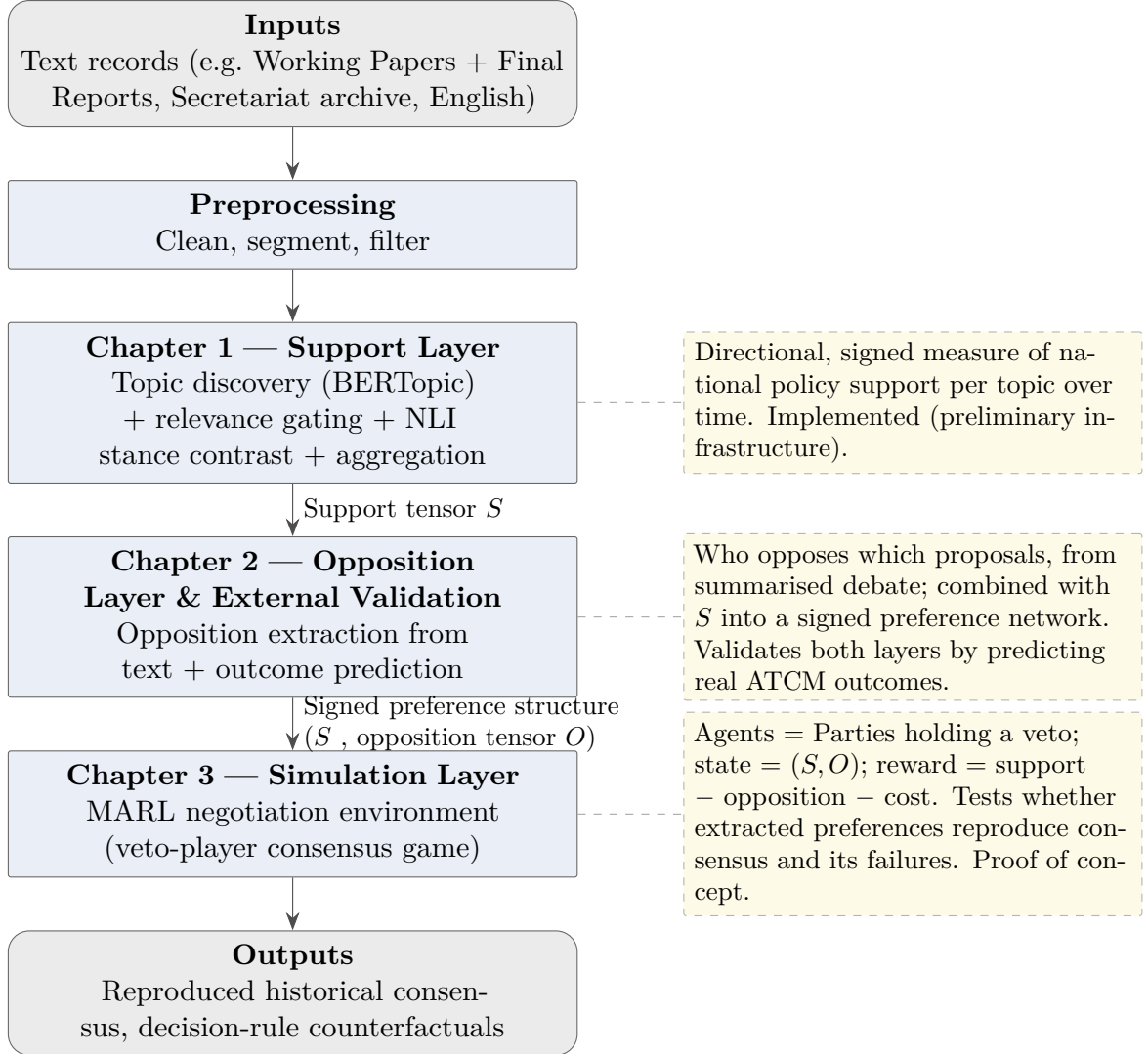


Figure 2.1: Thesis workflow: from text records to a validated consensus simulation.

Approach The approach is a pipeline with multiple stages (full details and notation in Appendix E). The first stages discover what the corpus is about: each document is embedded into the vector space introduced in Section 1.4 (detailed in Appendix G), and BERTopic (Grootendorst 2022) clusters the embedded corpus into a set of topics, each with a centroid: the average embedding of its documents. The next stages select, for each document and topic, the relevant passages, gated by cosine relevance to the topic centroid. The topic-relevant sentences are extracted and scored by the NLI classifier against topic hypotheses (e.g. country C supports stricter tourism rules), producing a five-class distribution that ranges from strong opposition to strong support. The signed stance of a document on a topic is the expected value of this distribution under an equal-spacing convention (Appendix E). It is a convex combination of five fixed class values, and therefore a continuous quantity between -1 and 1 on the topic’s axis.

Finally, the previous result is aggregated into the support tensor $\mathbf{S}$: for each country, topic, and year, the weighted sum of the stances of the WPs the country co-sponsored that year (Equation 3, Appendix E). This aggregation is a design choice of this thesis rather than a result from prior work: the attribution weights follow two conventions, full credit and fractional credit, each answering a different question (Appendix E). For a fixed year, the result is a signed country-by-topic matrix (full notation in Appendix C).

Measuring directional stance in diplomatic text is challenging because diplomats hedge: cautious, non-committal phrasing and a shared vocabulary of safeguards pull single-hypothesis scores toward neutrality (examples in Appendix E). Stance is therefore recovered as a contrast. For each topic, three hypotheses are written: a supporting pole, a neutral one, and an opposing pole (notation in Appendix E). Each is paired with the extracted text as premise and scored by the NLI classifier (Laurer et al. 2024), and the against-pole score is subtracted from the for-pole score, giving a direction that survives the hedged phrasing. As a robustness check, the same texts are scored a second time by a generative LLM used as a secondary instrument (Le Mens and Gallego 2025).

Validation strategy The pipeline above is an assertion built on stated assumptions: that meaning is geometric, that relevance is cosine similarity to a topic centroid, that stance is a contrast of entailment scores¹, and that co-sponsors share a paper’s position. None of these is guaranteed to hold on ATCM text. The construct is therefore validated with three independent checks, each targeting a different stage of the pipeline. A topic validation check compares the discovered topics against the categories the Secretariat assigns to every WP. Agreement is measured by majority-category overlap: the share of papers in each discovered topic that belong to that topic’s most common official category.

¹Entailment is the probability that the given premise supports the given hypothesis. Contradiction is the probability that it contradicts it.

A historical canary uses the minerals/CRAMRA negotiations of the late 1970s and 1980s as documented ground truth (Jackson 2021): most prominently, Australia and France abandoned the negotiated convention and pushed instead for a mining ban. The test checks whether the pipeline recovers those documented directions from the papers alone. Finally, a co-sponsorship check tests the assumption that countries co-sponsor a WP because they share its topic and stance: within the same year and topic, countries whose recovered stances are similar should be the ones that actually co-sponsor papers, with stance similarity computed excluding the paper being predicted.

2.1.2 Objective 2 / Chapter 2: The Opposition layer and external validation

Aim To measure not only what parties support but what they would block. Consensus fails through objection, so anticipating failure requires knowing who opposes what. This is the opposition signal: a directional measure of a party’s resistance to a proposal. It is not the inverse of the support layer: the support layer captures programmatic disagreement (a party proposes one direction while another proposes the contrary), whereas the *veto signal* is the act of objecting to a proposal on the table. Blocking is rarer and more consequential, and a party may hold the opposite stance on a topic yet never block. The signal has two sides: the veto signal against a given proposal, and a secondary, exploratory measure of a country’s general propensity to reject proposals within a given ATCM.

Approach As with the support layer, this chapter builds a pipeline. For the ATS case, the ATCMs’ final reports are the text input. Given the scarcity of text in which countries veto a proposal, the attribution is expected to be harder than for the support layer. An LLM extracts the statements in which a country objects to a proposal. Each extracted statement yields a *(country, target, polarity)* triple, with rationale and source quote. Taking the topics from Chapter 1 and mapping targets back to them yields an opposition tensor **O** with the same Country $\times$ Topic $\times$ Year structure, whose entries range from 0 (not likely at all to veto) to 1 (a certain veto). The pair **(S, O)** constitutes a signed preference network: the input to the third chapter.

External validation The combined structure (the tensor pair **(S, O)**) is tested for predictive content against observed outcomes (see Appendix F) and the compilation by Gardiner et al. (2025), for the case of the ATS. This gives Chapter 2 an independent research question, tests the results so far, and produces a quantitative description. Although this thesis focuses on the ATS, the pipeline is institution-agnostic: the same

(**S**, **O**) construction could be applied to other consensus- or veto-based international institutions whose deliberative records are available.

The main risks in this chapter are the sparsity of text that can be directly attributed to an input, and the identification of the country that vetoes any proposal. A further limitation is the influence of external geopolitics, as in the case of the consensus breakdown in 2022 due to the war in Ukraine (Gardiner et al. 2025). The cause of this type of exogenous influence is unlikely to be reflected in the text data of an institution not directly related to such issues.

Appendix D sketches the opposition layer pipeline and its external-validation step for the ATCM case study.

2.1.3 Objective 3 / Chapter 3: The Simulation layer

Aim To simulate consensus processes with country-agents whose state is parameterised by the empirically derived support and opposition from previous chapters. Its main measure of success is to be able to reproduce historical trajectories, and examine institutional counterfactuals. In consensus-based systems every Party is a veto player (Tsebelis 2002) and the representation is a strategic bargaining game over a status quo rather than an averaging dynamic.

Approach A custom negotiation environment is specified in which proposals are presented one at a time. Each proposal is an input text/Working Paper d that is mostly about topic k in year y . Then the agents (one per consultative party) decide whether to object. Each agent’s decision-making is supported by its support profile (its slice of **S**, Chapter 1), its opposition history (Chapter 2), and metadata such as consultative status, official language, and GDP. The metadata informs the agents’ decisions, but the thesis’s claims do not rest on it. An agent’s action on a presented proposal is binary: to object or not. Under the consensus rule the proposal is adopted only if no Party objects, so a single objection vetoes adoption. Each agent’s reward combines three terms: it rewards adoption of outcomes the Party supports, penalises adoption of outcomes it opposes, and subtracts a diplomatic cost for casting an objection (equations in Appendix E). Every agent holds a genuine veto, and consensus emerges from the trade-off between blocking a proposal and bearing the cost of objecting.

The model is validated by the predicted consensus or failure and by an ablation that removes different layers to quantify each layer’s contribution. Then a counterfactual analysis examines alternative decision rules, for example, qualified majority in place of unanimity, which relaxes the adoption rule (Equation 4, Appendix E). This coun-

terfactual is intended to predict the behaviour of the system given a change in the institutional rules, e.g. the ATCM framework (Secretariat of the Antarctic Treaty 2023). This is compared against the effectiveness metrics of Gardiner et al. (2025). Training agents on positions extracted from the institution’s own record makes this the highest-risk chapter; it is framed as a proof of concept, contingent on Chapters 1 and 2 passing validation.

2.2 Resources and funding required

The project uses openly published data from the Antarctic Treaty Secretariat’s document database and may use public data from other international institutions. The project uses open-source software and/or QUT-licensed software. Computation is served by the QUT high-performance computing cluster. No additional funding is required.

2.3 Individual contribution to the research team

I am the sole researcher on this project, with guidance from my supervisory team. The Chapter 1 infrastructure developed at the outset of candidature is preliminary work that establishes feasibility. This is not presented in this document as thesis results, and all reported findings will be produced within candidature. The thesis extends prior group work by Botelho et al. (2026), Kumar et al. (2026) and Carter et al. (2026).

2.4 Acknowledging use of Artificial Intelligence (AI) and AI-assisted technologies

I have used Claude Code for boilerplate and scaffolding code in the implementation of the pipelines and scripts. All architectural decisions, methodological choices, and algorithm design were specified by me in the input prompts. I also reviewed and edited generated code. I have also used Consensus AI and the Claude app to fetch the most relevant scientific papers from the available literature, based on my own ideas, keywords, and criteria. Afterwards, I read the papers and drew conclusions myself. Finally, I used Claude Code as a writing assistant for this document: rewording and help with grammar, readability, clarity and LaTeX document formatting. All research questions, project design, ideas, interpretation of results, and selection of literature are my own.

2.5 Timeline for completion

The candidature is full-time and commenced on the 31st of March, 2026. All milestones are placed against the QUT full-time schedule (Table [2.1](#)).

Table 2.1: Candidature timeline (April 2026–April 2029). ■ Ch. 1 ■ Ch. 2 ■ Ch. 3 ■ Milestones/events.



Table 2.1 (continued): Conference Attendance and Coursework.

Activity	3	6	9	12	15	18	21	24	27	30	33	36
	Jul	Oct	Jan	Apr	Jul	Oct	Jan	Apr	Jul	Oct	Jan	Apr
	'26	'26	'27	'27	'27	'27	'28	'28	'28	'28	'29	'29
Conference Attendance												
QANZIAM 2026												
AMEG Symposium 2026												
QANZIAM 2027												
International Conference												
AMEG Symposium 2027												
QANZIAM 2028												
Coursework												
RIO (completed)												
AIRS / IFN006												

Coursework (IFN006 Advanced Information Research Skills) is currently underway and will be completed by July 2026 following Stage 2 submission. The Research Integrity Online quiz is completed (see Appendix A).

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Appendices

Appendix A: Research Integrity Online

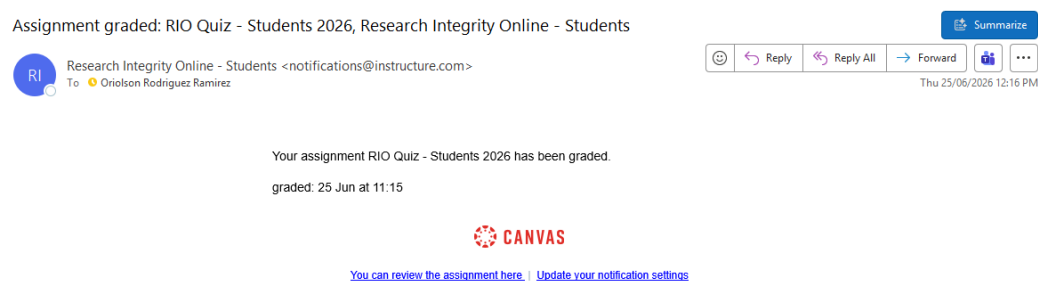


Figure 2: RIO completion notification email.

Grades for Oriolson Rodriguez Ramirez

Print Grades

Course: Research Integrity Online - ▾

Arrange by: Due date ▾

Apply

Name	Due	Submitted	Status	Score
RIO Quiz - Students 2026 RIO Quiz		25 Jun at 11:15		100 / 100
Assignments				
RIO Quiz				

Figure 3: RIO quiz completion record.

Appendix B: Coursework

AIRS / IFN006 is currently underway. The course ends in July 2026.

Appendix C: Notation

	c, i	countries (Consultative Parties), $c \in \{1, \dots, C\}$
	k	topic, $k \in K$; $ K $ is the number of topics
	y	year
	d	document (Working Paper)
	p	passage (chunk) of a document
Indices and sets.	ℓ	ordinal stance class, $\ell \in \{1, \dots, 5\}$
	$D(c, y)$	Working Papers co-sponsored by country c in year y
	$K(d)$	topics assigned to document d
	P_d	passages of document d , $P_d = \{p_1, \dots, p_{n_d}\}$
	$P_{d,k}$	passages of d selected as relevant to topic k
	A_d	co-sponsors of document d
	m	embedding dimensionality
	$\mathbf{e}_p$	embedding of passage p , $\mathbf{e}_p \in \mathbb{R}^m$
Embedding space.	$\boldsymbol{\mu}_k$	centroid of topic k , $\boldsymbol{\mu}_k \in \mathbb{R}^m$
	$\rho_{p,k}$	passage–topic relevance, $\rho_{p,k} = \cos(\mathbf{e}_p, \boldsymbol{\mu}_k)$
	τ	relevance selection threshold
	h_k^+, h_k^0, h_k^-	for-, neutral-, against-pole hypotheses for topic k
	$\lambda(d, h)$	null-premise-debiased entailment log-ratio
	$\tilde{\pi}_{d,k}$	three-pole distribution (stance classifier)
Stance and hypotheses.	$\pi_{d,k}$	five-class ordinal distribution, $\pi_{d,k} \in \Delta_5$
	Δ_5	probability simplex, $\{\pi \in \mathbb{R}^5 : \pi_\ell \geq 0, \sum_\ell \pi_\ell = 1\}$
	v	class values, $v = (-1, -\frac{1}{2}, 0, \frac{1}{2}, 1)$
	$s_{d,k}$	signed stance of d on topic k , $s_{d,k} \in [-1, 1]$
	$w_{d,c}$	attribution weight: 1 (full credit) or $1/ A_d $ (fractional)
Aggregation and tensors.	$S_{c,k,y}$	support of country c on topic k in year y , Eq. (3)
	$S_{\cdot, \cdot, y}$	year- y stance slice, $S_{\cdot, \cdot, y} \in \mathbb{R}^{C \times K }$
	$O_{c,k,y}$	opposition of country c on topic k in year y , $O_{c,k,y} \in [0, 1]$

For these tensors, high sparsity is expected because absence of evidence is recorded as missing, not as zero.

Appendix D: Opposition-layer workflow

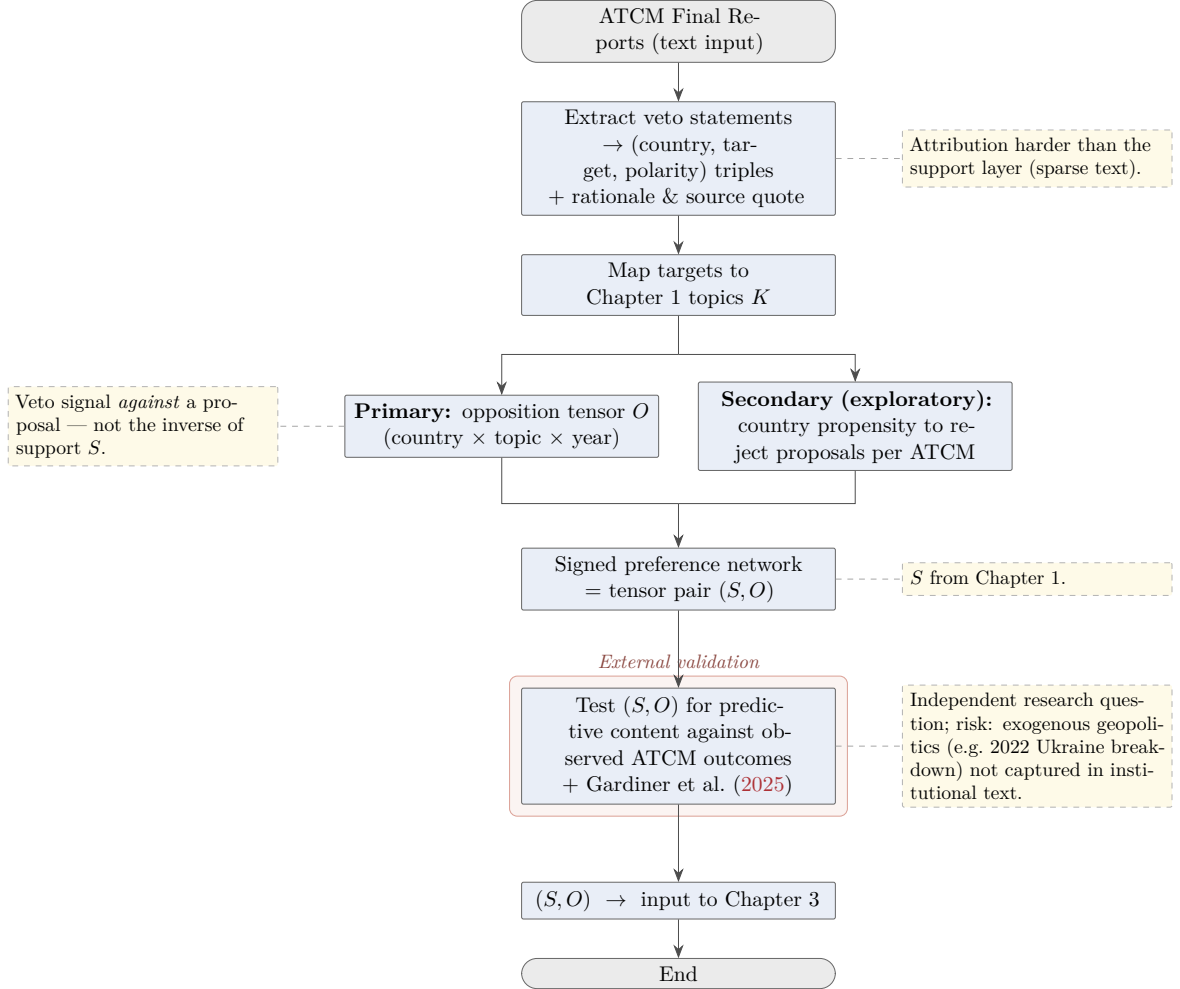


Figure 4: Chapter 2 opposition layer sketch: from text data to opposition tensor O , forming the signed preference network (S, O) .

Appendix E: Extended literature and method notes

ATCM document and output types. The ATCM’s inputs and outputs, including the paper types beyond Working Papers, are described in Appendix F.

Responsive (opinion-dynamics) model variants. Examples are the Kuramoto and opinion-changing-rate models used by Botelho et al. (2026), synchronization models that

represent agreement as a coupled-oscillator convergence, and the bounded-confidence (Hegselmann and Krause 2002) and stubborn-agent (Friedkin and Johnsen 1990) variants of the iterative-averaging scheme of DeGroot (1974).

Veto-players condition (Tsebelis 2002). Let q denote the status quo, p a proposal, and u_i the utility function of party i , which scores how much that party values each policy. The status quo is replaced only when $u_i(p) \geq u_i(q)$ for every party i : no party may be left worse off than it is now.

Foundational MARL literature. Foundational MARL work covers multi-issue bargaining (Chang 2021), negotiating-team formation (Bachrach et al. 2020), bargaining under risk and asymmetric information (Schmid et al. 2020), and search-improved game-theoretic learning (Li et al. 2023). In these systems, neural networks (NN) enter only as function approximators, allowing the learning to scale to large state and action spaces.

LLM-based negotiation simulations. Moghimifar et al. (2024) simulate coalition formation in parliament using text data from party manifestos. Bolleddu (2025) present a MARL system for consensus building and negotiation based on synthetic data. Kulkarni et al. (2025) use negotiated agreements with game-theoretic reasoning to predict dynamic coalition formation in Diplomacy, a board game in which seven powers negotiate, form, and break alliances, widely used as a benchmark for this type of simulation. Some benchmarks for real negotiation data are starting to appear (Benac et al. 2026).

Wordfish (Slapin and Proksch 2008). Its aim was to estimate party positions over time without hand-coded reference texts. The model treats each word count as a Poisson draw whose expected frequency depends on the party’s position on a single latent dimension, and it recovered credible left–right trajectories for German parties from their election manifestos (1990–2005).

Wordshoal (Lauderdale and Herzog 2016). Their model scales each debate separately with Wordfish and then combines the debate-level positions into a general dimension through a Bayesian factor analysis. Applied to speeches in the Irish Dáil and the US Senate, it recovered within-party disagreement that voting records hide. Each debate scale is whatever axis dominates that debate’s word frequencies, which need not be the substantive for-or-against axis of interest. Lauderdale and Herzog get away with this because legislative debates are adversarial, so the dominant axis is usually government versus opposition; ATCM Working Papers have no such structure.

BERTopic and NLI models. BERTopic (Grootendorst 2022) clusters document embeddings and labels each cluster with its most distinctive words. Natural language inference (NLI) models (Burnham 2023; Abercrombie and Batista-Navarro 2022) are classifiers that take a text and a stated hypothesis (e.g. “this paper supports stricter protection”) and judge whether the text entails, contradicts, or is neutral towards it. A single stable ideal point is too coarse for the ATCM, where a party can support stricter environmental protection yet oppose new tourism rules.

Support-layer pipeline details. Each document is embedded: mapped to a vector in $\mathbb{R}^m$, under the geometric assumption that documents with similar meaning receive nearby vectors. BERTopic reduces the dimension of these vectors and clusters them, and each cluster is a topic; each centroid $\mu_k \in \mathbb{R}^m$ is the average embedding of the documents in topic k . Each document d is split into passages $P_d = \{p_1, \dots, p_{n_d}\}$, and each passage p is embedded as $\mathbf{e}_p \in \mathbb{R}^m$. Relevance is defined quantitatively: $\rho_{p,k} = \cos(\mathbf{e}_p, \mu_k)$, the cosine similarity between passage embedding and topic centroid, so a passage that points in the same direction as the topic centroid discusses that topic. Relevance acts as a gate: $P_{d,k}$ is the set of passages scoring above a threshold, and only these pass to the next stage. The topic-relevant sentences in each $P_{d,k}$ are extracted using a Large Language Model (LLM); $\text{Extract}(P_{d,k})$ denotes this step. The extracted sentences are concatenated and scored as a set by the NLI classifier: given the text as premise and a hypothesis about topic k , it scores entailment, neutrality, and contradiction. Entailment means the premise gives support to the hypothesis. Contradiction means the premise contradicts it. The resulting scores are binned and combined into five ordinal classes,

$$\pi_{d,k} = \text{NLI}(\text{Extract}(P_{d,k})) \in \Delta_5, \quad \Delta_5 = \left\{ \pi \in \mathbb{R}^5 \mid \pi_\ell \geq 0, \sum_{\ell=1}^5 \pi_\ell = 1 \right\}, \quad (1)$$

where Δ_5 is the probability simplex: $\pi_{d,k}$ assigns a non-negative probability to each of the five classes, and these probabilities sum to one. The signed stance is the expected value of this distribution under an equal-spacing convention,

$$s_{d,k} = \sum_{\ell=1}^5 v_\ell \pi_{d,k}(\ell), \quad v = \left(-1, -\frac{1}{2}, 0, \frac{1}{2}, 1 \right). \quad (2)$$

Since $\pi_{d,k}$ is a probability distribution and the five fixed class values v_ℓ all lie in $[-1, 1]$, the stance $s_{d,k}$ is a convex combination of those values. The stance is therefore a continuous quantity in $[-1, 1]$ on topic k ’s axis, while v itself never varies.

Stance classifier (Chapter 1). For topic k , let h_k^+, h_k^0, h_k^- be the for-, neutral-, and against-pole hypotheses. For passage set d and hypothesis h , the debiased evidence is

the null-premise-corrected log-ratio,

$$\lambda(d, h) = \text{clip}\left(\log P(\text{entail} \mid d, h) - \log P(\text{entail} \mid \emptyset, h), \pm 5\right),$$

this removes each hypothesis’s marginal acceptability. The three-pole distribution combines entailment of each pole with contradiction of its opposite,

$$\tilde{\pi}_{d,k} = \text{softmax}\begin{pmatrix} \lambda_{\text{ent}}(h_k^+) + \lambda_{\text{con}}(h_k^-) \\ \lambda_{\text{ent}}(h_k^0) \\ \lambda_{\text{ent}}(h_k^-) + \lambda_{\text{con}}(h_k^+) \end{pmatrix},$$

these values are reduced to a continuous score in $[-1, 1]$: the against-pole probability subtracted from the for-pole probability. The score is binned into the five ordinal classes of $\pi_{d,k}$ at the midpoints of the class values v : scores in $[-1, -0.75)$ are strongly against, $[-0.75, -0.25)$ against, $[-0.25, 0.25)$ neutral, $[0.25, 0.75)$ in favour, and $[0.75, 1]$ strongly in favour. The scalar stance is then the expected value of Equation 2.

Aggregation (Chapter 1). The document stances are aggregated over the Working Papers, weighted by co-sponsorship credit,

$$S_{c,k,y} = \sum_{\substack{d \in D(c,y) \\ k \in K(d)}} w_{d,c} s_{d,k}, \quad (3)$$

where $S_{c,k,y}$ is the support of country c on topic k in year y , $D(c, y)$ is the set of Working Papers co-authored by country c in year y , $K(d)$ is the set of topics assigned to document d , and $w_{d,c}$ is the attribution weight of that country in one document. For a fixed year y , the slice $S_{\cdot, \cdot, y} \in \mathbb{R}^{C \times |K|}$ is a signed country $\times$ topic matrix, with C countries and $|K|$ topics; a dot means the index runs over all its values.

Attribution weights (Chapter 1). Two attribution conventions are computed:

- Full credit, $w_{d,c} = 1$: every co-sponsor fully holds the position. This is the coalition-membership reading (“who is on this side?”).
- Fractional credit, $w_{d,c} = 1/|A_d|$, with A_d the co-sponsors of d : the paper’s positional mass is divided among its sponsors, discounting states that merely sign on. This is the influence reading (“who is driving this?”).

Hedging in diplomatic text. Hedging means cautious, non-committal phrasing: a party rarely writes “we oppose this measure”, it writes “further consideration may be

warranted”. Positions on opposite sides also share the same vocabulary of safeguards: a phrase such as “subject to appropriate environmental safeguards” appears in permissive and restrictive papers alike. Example pole hypotheses for a topic: h_k^+ “tourism should be further regulated”, h_k^- “tourism regulation should be relaxed”.

Other ideal-point precedents. Vafa et al. (2020) produced a probabilistic framework that jointly places ideal points and models actors’ language.

Embeddings. An embedding may cover a word, a sentence, or a whole document. The map is learned from large corpora so that texts appearing in similar contexts receive nearby vectors. Neural language models such as BERT (Devlin et al. 2019) make the vectors context-dependent: the same word receives different vectors in different sentences.

The UN General Debate corpus (Baturo et al. 2017). Their corpus collects over 7300 annual General Debate statements (1970–2014), one per country per year, in a setting that parallels the ATCM: formal, annual, state-level, and translated into official languages. They derive preferences in two ways, both from word frequencies: Wordscores, a supervised scaling in which the analyst defines the axis in advance through reference texts (US and Russian statements anchor a US–Russia dimension), and correspondence analysis, an unsupervised reduction whose dimensions are interpreted afterwards.

Opposition tensor (Chapter 2). $O_{c,k,y} \in [0, 1]$ is the opposition of country c on topic k in year y , built from the extracted *(country, target, polarity)* triples: 0 means the country is not likely at all to veto, and 1 means a certain veto.

Simulation-layer equations (Chapter 3). Each agent’s action on a presented proposal is binary, $o_i \in \{0, 1\}$: one if agent i objects, zero otherwise. Under the consensus rule the proposal is adopted only if no Party objects,

$$1_{\text{adopt}} = \prod_j (1 - o_j), \quad (4)$$

therefore, a single objection vetoes adoption. The reward to agent i is then

$$R_i = \alpha S_{i,k,y} 1_{\text{adopt}} - \beta O_{i,k,y} 1_{\text{adopt}} - \gamma o_i, \quad (5)$$

where the first term rewards adoption of outcomes the Party supports, the second penalises adoption of outcomes it opposes, the third subtracts a diplomatic cost for

casting an objection, and $\alpha, \beta, \gamma \geq 0$ are trade-off weights, learned during training to fit the historical consensus record.

Appendix F: The ATS as a consensus regime

At the ATCM, Consultative Parties (CPs) reach decisions exclusively by consensus; non-consultative parties (NCPs), observers, and experts attend but take no part. These outputs take three forms (Secretariat of the Antarctic Treaty 2023, Rule 24). **Measures** set out substantive provisions that the parties intend to be legally binding in their domestic law; they require each party's government to formally approve them. **Decisions** settle the ATCM's own internal and organisational matters. **Resolutions** are non-binding: they record recommendations or expressions of intent, with no legal force.

There are four types of input. The one that matters for this thesis is the **Working Paper (WP)** (Secretariat of the Antarctic Treaty 2023, Rule 48): submitted by CPs and Observers, WPs require discussion at the ATCM and are the main source of outputs. They are the only input from CPs that can produce an outcome. **Secretariat Papers** (Secretariat of the Antarctic Treaty 2023, Rule 49) cover organisational matters such as the budget. **Information Papers** (Secretariat of the Antarctic Treaty 2023, Rule 50) provide supporting information, also from NCPs and experts. **Background Papers** (Secretariat of the Antarctic Treaty 2023, Rule 51) are documents submitted for the record only.

The consensus system makes the ATS a notable example of international cooperation (Jackson 2021); however, the same mechanism has become a major challenge as the number of CPs grows (Tsebelis 2002; Gardiner et al. 2025).

A notable case of contested consensus occurred in the late 1970s and 1980s, during the negotiations over mineral resources in Antarctica. The Convention on the Regulation of Antarctic Mineral Resource Activities (CRAMRA) was a framework, agreed in 1988, that would have permitted and regulated mining in Antarctica. It was abandoned before entering into force and replaced by the Madrid Protocol (1991), which instead prohibited all mineral resource extraction and designated Antarctica a natural reserve devoted to peace and science (Jackson 2021). Although it unfolded in two stages, it forms a single documented episode: one negotiation over Antarctic mineral resources in which the parties' positions are recorded and then reversed. That documented reversal makes it a valuable canary test, a known case against which any quantitative stance measure can be validated.

Appendix G: Measuring policy preference from political text

Positional measures place each actor on a policy scale, for example from favouring regulation to favouring access, using only the text the actor produces. Extracting them from political texts is a well-developed field with two paradigms. They differ in how they turn text into numbers: the first paradigm uses word counts (i.e. frequencies), while the second uses embeddings. The first paradigm is scaling and ideal-point methods, which recover a latent positional dimension from the data itself: the axis is not chosen in advance, it emerges from patterns in the text, and its meaning is interpreted after estimation. Wordfish (Slapin and Proksch 2008) is a Poisson scaling model of party text (Appendix E). Its limitation is that the recovered dimension is uninterpreted: it is the dominant axis of variation in word usage, and in a corpus as topically diverse as the ATCM Working Papers it would likely reflect what parties talk about rather than what they support. Lauderdale and Herzog (2016) addressed the problem that legislators talk about different things in different debates by scaling each debate separately, but the debate scales inherit the same uninterpreted-dimension problem, and the debates must be given in advance (Appendix E). Two things are therefore missing for this thesis: the topics have to be discovered from the papers themselves, and the output must be a stance per topic rather than one general dimension. The closest dataset to the ATCM corpus is the UN debate corpus Baturo et al. (2017) use to derive state-level preferences, in a setting that parallels the ATCM (Appendix E). Finally, Parschan and Jakob (2025) review the text-based ideal-point estimation algorithms of this first paradigm, comparing their assumptions and variants in one place. The second paradigm uses embeddings as its main tool. An embedding maps a piece of text to a vector in $\mathbb{R}^d$ so that texts with similar meaning receive nearby vectors: comparisons of meaning become geometric operations. Modern embeddings are produced by neural language models such as BERT (Devlin et al. 2019) (details in Appendix E). Word-embedding geometry recovers ideological placement in parliamentary data (Rheault and Cochrane 2019), but the choice of embeddings strongly affects applied inference and the results must be validated (Rodriguez and Spirling 2022).

This thesis adopts the second paradigm and uses embeddings at several stages: to discover what topics the Working Papers discuss, to measure how relevant each paper is to each topic, and, through the same family of neural models, to read each paper’s stance on a topic. This yields the topic-conditioned stance that the first paradigm cannot supply: a separate position per party and topic, allowed to change through time. The topic structure is discovered using BERTopic (Grootendorst 2022) and the stance is read with natural language inference (NLI) models (Burnham 2023; Abercrombie and Batista-

Navarro 2022), both described in Appendix E. Section 2.1.2 sets out the validation checks for these results: against the Secretariat’s hand-assigned topic categories, against the documented CRAMRA reversal, and against observed co-sponsorship.